

Development of an Audio Device with Steerable Sound Beams Using Acoustic Lens Principles

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Part 1: Theoretical Foundation and Simulation Methodology

Introduction: The Concept of an "Audio Spot"

In the same way an optical lens can refract light to a focal point, an **acoustic lens** can bend the path of sound waves. This principle allows for the creation of highly directional sound, forming an "audio spot"—a confined area where sound is clearly audible, while remaining quiet just a short distance away. This technology opens up possibilities for personalized audio experiences in public spaces, museums, or even open-plan offices, delivering sound only to the intended listener.

This report details an experiment that simulates the structure of an acoustic lens using a one-dimensional array of speakers. By introducing precise time delays (phase shifts) to the signal of each speaker, we can control the direction of the resulting sound wave. The objective is to simulate this phenomenon, analyze its characteristics, and verify if a steerable audio beam can be effectively formed.

Principle of Operation: Phased Array Speakers

The core of this experiment lies in the concept of a **phased array**. Imagine a line of speakers placed at equal intervals. If all speakers emit sound simultaneously, the sound wave propagates perpendicularly from the array. However, if we introduce a small, progressive time delay to each subsequent speaker, the wavefronts will combine and interfere constructively in a specific direction, effectively "steering" the sound beam.

The direction of the sound beam can be calculated based on the speaker spacing, the time delay, and the speed of sound.

The relationship is described by the following formula:

$$\tan(\theta) = (v * s) / d$$

Where:

- θ is the angle of the sound propagation direction.
- v is the speed of sound (approx. 340 m/s).
- s is the time delay between adjacent speakers.
- d is the distance between adjacent speakers.

This experiment uses a simulated array of nine speakers to test this principle, varying parameters such as frequency, time delay, and speaker spacing to compare the measured beam angle with the theoretical angle calculated from the formula.

Simulation Methodology

The experiment was conducted using [EASE Focus 3](#), a professional acoustic simulation software. This tool allows for the creation of virtual spaces and the configuration of sound sources with specific properties. For this study, a one-dimensional array of nine audio sources was modeled, as depicted in the principle diagram. The simulation measured the spatial characteristics of the sound field to determine how the audio beam formed and propagated under different conditions.

Part 2: Experimental Results and Future Prospects

Experimental Results

The simulations demonstrated that by adjusting the phase delay between speakers, the direction of sound propagation could be effectively controlled. For example, with a fixed speaker spacing of 5 meters and a frequency of 1 kHz, increasing the time delay from 1ms to 3ms clearly shifted the angle of the primary sound lobe. The results confirmed that at a sufficient distance from the speaker array, a highly directional beam—an audio spot—is formed, propagating over a long distance at a specific angle.

A key finding was the strong correlation between the theoretical calculations and the simulated results, particularly when the phase delay was relatively small. This validates the underlying principle of using phased arrays to steer sound.

Detailed Analysis of Beam Steering

To quantify the performance, the measured angle ($\tan\theta$) was plotted against changes in phase delay, frequency, and speaker spacing.

Impact of Phase Delay on Beam Angle

As shown in the chart below, increasing the phase delay (s) results in a wider steering angle (larger $\tan\theta$). The measured values closely follow the theoretical predictions, especially for delays under 2ms. At higher delays, a slight deviation is observed, but the overall trend is consistent. This demonstrates that the beam can be steered across a wide range; the observed change in $\tan\theta$ up to approximately 1.4 corresponds to an angle of about 54 degrees, enabling a total steering range of over 100 degrees.

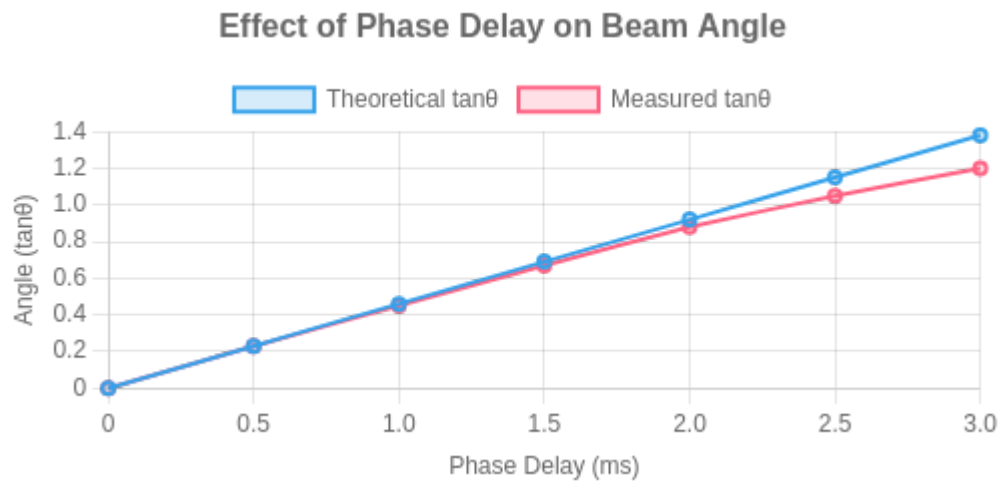


Figure 1: Relationship between Phase Delay and Beam Angle ($\tan\theta$)

Impact of Frequency and Spacing

The theoretical formula suggests that the steering angle is independent of frequency. The simulation results confirm this, showing that $\tan\theta$ remains nearly constant across the tested frequency range (0.5 kHz to 3.0 kHz). Conversely, the angle is inversely proportional to the speaker spacing (d). As the distance between speakers increases, the steering angle for a given time delay decreases, a behavior that was also accurately reflected in the simulations.

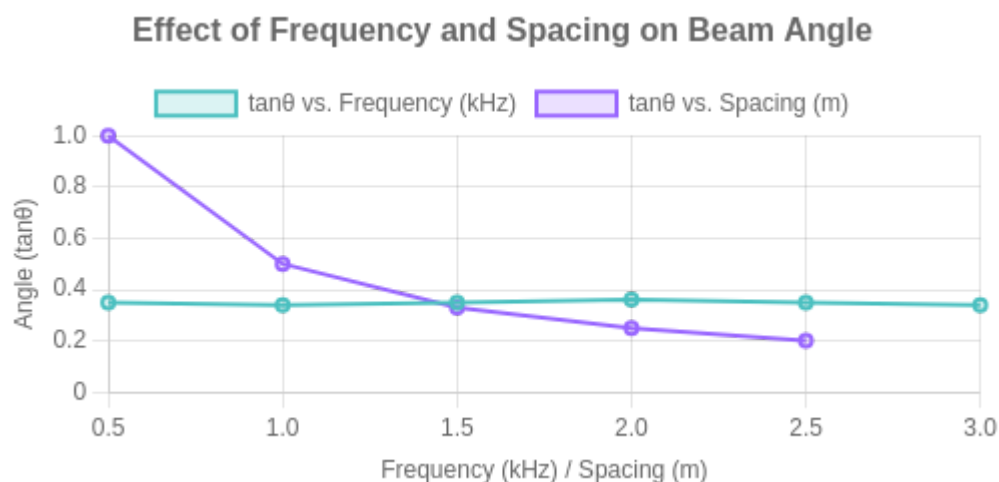


Figure 2: Impact of Frequency and Speaker Spacing on Beam Angle ($\tan\theta$)

Conclusion and Future Prospects

This study successfully demonstrated that an "audio spot" can be formed and steered using a phased array of speakers, achieving effects similar to a physical acoustic lens. The beam's directionality was confirmed to be controllable and predictable, especially at a reasonable distance from the source and with smaller phase delays.

The key takeaway is that phased array technology provides a viable method for creating steerable, directional audio. However, limitations exist, particularly with audible frequencies which exhibit lower directivity, especially in close proximity to the speakers.

A promising direction for future work is the use of **ultrasonic speakers**. Higher frequencies naturally produce more focused beams of sound. A parametric speaker, for instance, uses modulated ultrasonic waves. These waves interfere with each other in the air, and this non-linear interaction demodulates the signal, generating audible sound only along the narrow path of the ultrasonic beam. This creates an extremely precise audio spot.

This approach could be further enhanced using Frequency Division Multiple Access (FDMA). By assigning different ultrasonic carrier frequencies to different audio channels, each with its own phase delay, it would be theoretically possible to create **multiple, independently steerable audio spots from**

a single speaker array. This would allow for complex, multi-user audio environments powered by a single, compact device, unlocking a wide range of new applications.

References

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